

Enabling HPC Earthquake and Tsunami Simulation Data Mining: an NFDI4Earth Lighthouse Use Case Project*

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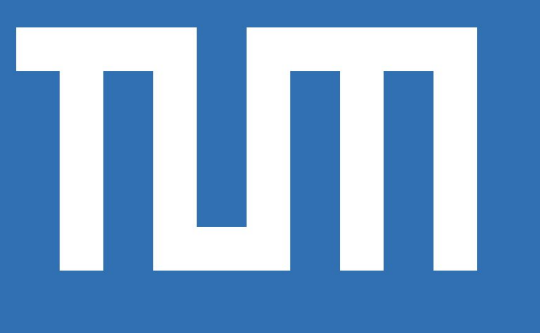


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Preparatory Work

**“High-Performance Computing Workflow for Rapid-Response
Physics-Based 3D Earthquake Simulations Integrated in Data Lake”**
ISC 2026 poster by Iris Christadler, Mathilde Marchandon and Alice-Agnes Gabriel showing a seismic workflow that utilizes a pre-computed catalog of numerical simulation results from SeisSol for Alto Tiberina.

Motivation

Physics-based 3D earthquake and tsunami simulations require vast HPC resources. But once results are published, data are often lost in scattered repositories (Zenodo, Github, ...) with little standardization. This project aims to make simulation data FAIR (Findable, Accessible, Interoperable, Reusable) and usable for rapid scientific and societal applications.

Objective

Unify research data management (RDM) for large-scale Earth science simulations through open standards and HPC-ready workflows.

1. Develop and promote international **standards for simulation (meta-)data formats** for large-scale earthquake/tsunami simulations
2. **Develop tools** to access, combine and post process simulation and observational datasets
3. **Provide derived, labeled products** for downstream AI data mining and use in scientific and civil protection use cases

Vision

Rapid, easy access to standardized simulation results through:

- Metadata schema + wizard: Structured FAIR upload
- Simulation Data Lake: Persistent HPC data storage
- Data tools: Derived data products for analysis and AI workflows

Methodology

- Design metadata schema based on input and results of numerical simulations
- Implement metadata wizards for automatic data upload
- Establish a Simulation Data Lake with FAIR data access
- Integrate visualization & data tools (e.g., shake-maps)

Why a Lighthouse Use Case?

As an NFDI4Earth Lighthouse Use Case project, this initiative demonstrates how standardized, machine-readable metadata and shared workflows can:

- Reduce duplication and increase reproducibility,
- Foster cross-disciplinary data mining, and
- Accelerate integration between earthquake and tsunami simulation codes

Use Cases

- Large-scale 3D dynamic rupture modeling (SeisSol)
- Combined observational + synthetic data mining
- AI training using labeled seismic products

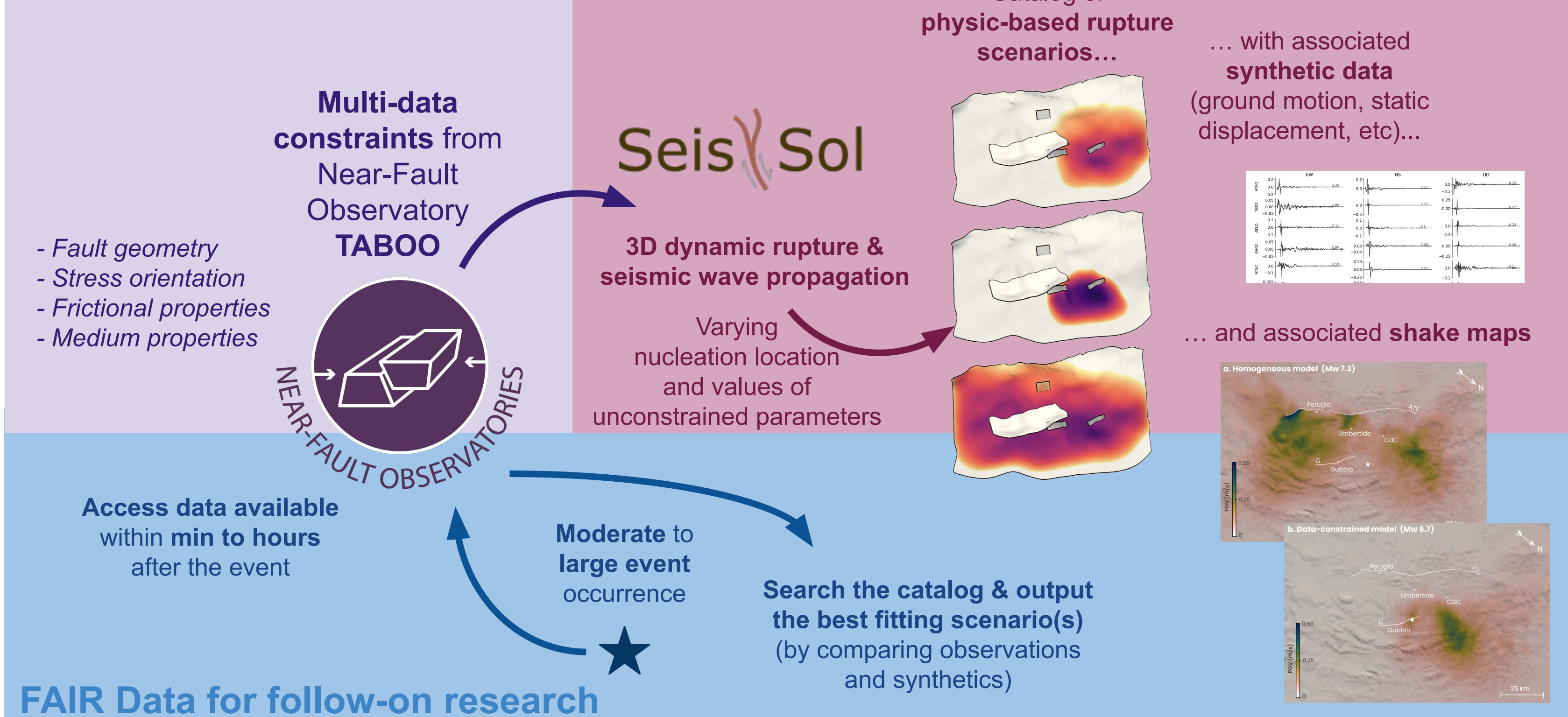
Target Users

Earth System Scientists, Research Software Engineers, HPC users

Engage with us

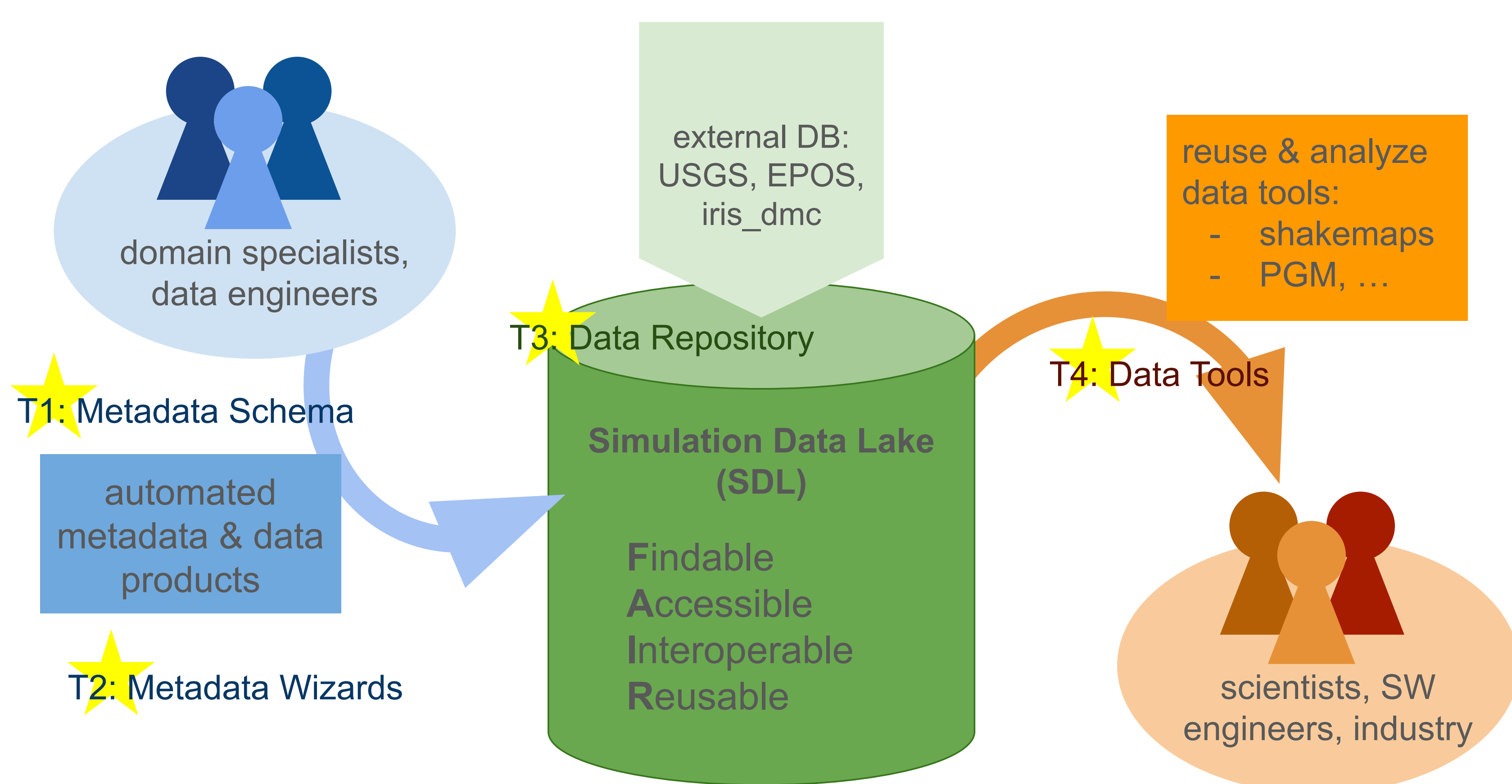
We welcome scientific exchange (on seismic metadata and data hosting solutions) and early adopters for the repository pilot phase. Please get in contact with iris.christadler@lmu.de

Observation data & physics



Vision

Rapid, easy access to standardized simulation results



Project Overview

Year 1: Setup

(Oct.26-Sep.27)

- Develop metadata schema
- Prototype metadata wizard
- Prepare data lake
- Initial community training

Year 2: Integration

(Oct.27-Sep.28)

- Train early users
- First data ingestion
- Refine schema & wizard
- Initial tool integration

Year 3: Operation

(Oct.28-Sep.29)

- Datathon > Community use
- Operational data lake
- Connect data tools
- Automate FAIR workflows

References

Mathilde Marchandon et al. “Forecasting 3D Rupture Dynamics of the Alto Tiberina Low-Angle Normal Fault, Italy” In: *Seismica* 4.2 (2025), doi:10.26443/seismica.v4i2.1603
Ulrich, T., Gabriel, A.-A. and Madden, E. H. “Stress, rigidity and sediment strength control megathrust earthquake and tsunami dynamics” In: *Nature Geoscience*, 15, 67-73, (2022) doi:10.1038/s41561-021-00863-5
Lukas Krenz et al. “3D acoustic-elastic coupling with gravity: the dynamics of the 2018 Palu, Sulawesi earthquake and tsunami”. In: *Proceedings of the International Conference for High-Performance Computing, Networking, Storage and Analysis. SC '21. St. Louis, Missouri: ACM, 2021. ISBN: 9781450384421. doi:10.1145/3458817.3476173*

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